TASK SPECIFICATION: P1-03 PHASE P1 — FOUNDATIONS & DEVOPS INFRASTRUCTURE

Backend Core Modular Monolith Skeleton (Spring Boot 3.3, Java 21 LTS)

Git Branch: `feature/P1-03-backend-skeleton` · Target Pull Request: PR (feature/P1-03-backend-skeleton)

Core Objective & Technical Scope:

Bootstrap Spring Boot 3.3 modular monolith with Java 21 LTS virtual threads, Actuator telemetry, SpringDoc OpenAPI/Swagger, RFC 7807 ProblemDetail exception handling, and ArchUnit architecture governance.

Priority Level	High	Complexity Rating	`M`
Execution Status	Done ✅	Target PR / Commit	`PR (feature/P1-03-backend-skeleton)`
Upstream Dependencies	P1-01, ADR-0001		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Initialize Maven multi-module structure with Java 21 and Spring Boot 3.3.4.
2. Implement GlobalExceptionHandler emitting RFC 7807 ProblemDetail responses.
3. Configure SpringDoc OpenAPI with Swagger UI at /swagger-ui/index.html.
4. Configure application profiles (dev, test, prod) with isolated configurations.
5. Define and enforce ArchUnit architecture rules preventing illegal cross-module dependencies.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/pom.xml`
- `backend/src/main/java/.../common/**`
- `backend/src/main/resources/application*.yml`
- `backend/src/test/**`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Backend starts successfully on port 8080 with Actuator health endpoint UP.
- Swagger UI loads at /swagger-ui/index.html.
- Validation failures return standard RFC 7807 problem responses with localized error messages.

VERIFICATION & TEST COVERAGE STRATEGY

- GlobalExceptionHandlerTest
- ArchUnitTest
- ContextLoadTest
- Maven verify